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United States Patent	12418455
Kind Code	B2
Date of Patent	September 16, 2025
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Merging configurations of network devices

Abstract

Systems and methods for merging configurations on network devices based on structured models representing features to be configured are disclosed. By utilizing these structured models embodiments may generally allow configuration merges to be performed substantially on a per feature basis and also allow the merging to be performed in a manner that minimizes conflicts and implementation complexity while substantially improving the performance of those configuration merges.

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Appl. No.: 18/071826

Filed: November 30, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240179057 A1	May. 30, 2024

Publication Classification

Int. Cl.: H04L41/082 (20220101); H04L41/0853 (20220101); H04L41/0859 (20220101)

U.S. Cl.:

CPC H04L41/082 (20130101); H04L41/0856 (20130101); H04L41/0863 (20130101);

Field of Classification Search

CPC: H04L (41/082); H04L (41/0856); H04L (41/0863)

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Background/Summary

BACKGROUND

(1) The configurations of network devices which control the flow of packets in a network, such as a router or switch, may be quite intricate, including thousands (or more) of individual features. Typically the configuration of these individual features may be accomplished through interaction with an interface of the network device. A user can input commands for configuring one or more features through the interface and the features may be configured accordingly at the network device.

(2) As may be imagined, configuring such network devices can be time consuming and complex. Additionally, certain problems related to issues of synchronization between various configurations of the network device may arise. For example, in most cases there are no restrictions on the number of users who may perform such configurations or the timing of such configurations. Moreover, there is usually no effective way of reconciling these configuration changes. Accordingly, configurations by one user may overwrite, or otherwise replace or negate, changes to the configuration made by another user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings accompanying and forming part of this specification are included to depict certain aspects of the disclosure. It should be noted that the features illustrated in the drawings are not necessarily drawn to scale. A more complete understanding of the disclosure and the advantages thereof may be acquired by referring to the following description, taken in conjunction with the accompanying drawings in which like reference numbers indicate like features.

- (2) FIG. 1A is a block diagram of a network system including an embodiment of a network device for merging configurations.
- (3) FIG. 1B is a block diagram of an embodiment of a network device for merging configurations.
- (4) FIG. 1C is a block diagram of a control plane of embodiments of a network device for merging configurations.
- (5) FIG. 2 is a block diagram depicting a general architecture of a network device for merging configurations.
- (6) FIG. 3 is a block diagram depicting an example of an embodiment of a network device for merging configurations.
- (7) FIGS. 4A and 4B are a flow diagram of an embodiment of a method for merging configurations.

DETAILED DESCRIPTION

- (8) As discussed, the configurations of network devices, such as a router or switch, may be quite intricate, including a large number of individual features (e.g., parameters, variables, etc.). At a fundamental level the configuration of these individual features may be accomplished through interaction with an interface, such as a command line interface (CLI) or the like. A user can thus input a command specifying a value for one or more features through the interface and the features may be configured accordingly at the network device.
- (9) As may be imagined, configuring such network devices can be time consuming and complex, as individual commands must be entered in order to (re)configure the myriad features of the network device. Additionally, complicating such configuration is the fact that many of the interfaces for configuration of a network device are what are termed “additive,” meaning that if it is desired to remove a feature (e.g., a command) from a running (e.g., existing) configuration, a command usually needs to be made to negate the existing feature.
- (10) Given these methods and limitations for configuration, the configuration of features of a network device is complex and time consuming, especially where a large number of features are involved. Moreover, in some cases the only way to reconfigure a device completely (or substantially) is either to carefully negate, or replace, every existing command (a practical impossibility in most circumstances), or to overwrite a startup configuration and reload the network device. As may be imagined, this is a massively disruptive process that can take a large amount of time depending on the device and configuration desired.
- (11) To aid in the configuration of network devices and attempt to ameliorate these issues, providers (e.g., manufacturers or software providers) of these network devices introduced configuration sessions. A configuration session allows configuration changes to be made in a temporary location. The configuration session can then be “committed” to replace a configuration running on the network device (the “running” or “current” configuration) with the configuration defined in the session (the “session” configuration).
- (12) A configuration session may include an ancestor configuration and a session configuration. To illustrate in more detail, when a configuration session is initiated the configuration session may be assigned a unique name and associated with a memory location or space. At the time the configuration session is initiated, the current configuration running on the device at that time may be referenced in the configuration session as the “ancestor” or “initial running” configuration with respect to that configuration session. If the current running configuration is modified subsequently to the initiation of that configuration session, a copy (or snapshot) of (at least the modified portion of) the current running configuration (e.g., at the time the configuration session was initiated) can then be saved as the ancestor configuration in the memory location associated with the configuration session. For example, if (at least a portion) of the configuration is modified during the configuration session, the corresponding (portion of) the current running configuration can be copied to the ancestor configuration in the memory location associated with the configuration session.

(13) The configuration session thus allows configuration changes to be made to a configuration (e.g., the configuration at the time the configuration session was initiated as included in the ancestor configuration) in a session configuration (e.g., configuration changes in the session configuration are relative to the ancestor configuration associated with the initial running configuration) without altering the actual running configuration of the network device. In other words, configuration commands may be executed against this session configuration to alter this session configuration using a configuration interface (e.g., a CLI) without changing the running configuration of the network device. Thus, a configuration session can be entered, modified, or exited at any time without impacting a currently running system configuration.

(14) At some point then, a user may commit the session configuration (e.g., when the user is done entering configuration commands). When the session configuration is committed (e.g., a user indicates that they wish to configure the network device according to the configuration session), the currently running configuration on the network device is replaced with the session configuration, which includes any changes in configuration made during the configuration session (e.g., any configuration commands entered as part of the configuration session) relative to the ancestor configuration (the running configuration at the time the session was initiated).

(15) While the use of these types of configuration sessions may alleviate some of the problems with respect to configuring devices, these methods of device configuration are not without problems of their own. In the main, these problems relate to issues of synchronization between various configurations of the network device. Namely, in most cases implementations of configuration sessions overwrite the current configuration of the network device with the session configuration at the time the configuration session is committed. Thus, any changes that were made to the running configuration on the network device between the time a configuration session was initiated and when the configuration session was committed, may be lost. If different configuration sessions modify different features, the later session might override changes from an earlier session, and if a rollback (e.g., to a previous configuration state) is performed, it may override the configuration changes from a previous session.

(16) These issues are exacerbated by the fact that many providers of network devices allow multiple configuration sessions to be ongoing simultaneously, while also allowing configuration changes to be made (e.g., in real time) through the regular configuration interface on the device itself. Accordingly, when a configuration session is committed, any changes made to the running configuration on the device between the time the configuration session was initiated and the time the session was committed will be overwritten, and thus lost. Additionally, from a user standpoint, no alert is given to a user (e.g., an administrator, a user who initiated a configuration session, etc.) that configuration changes may be overwritten. This is typically undesirable behavior.

(17) What is desired then, is to allow configuration changes to be merged into a currently running configuration and to determine when such configuration changes may conflict to provide a user with such information.

(18) To illustrate in more detail, it is both desired to minimize conflicts—the configuration of different features, or the same feature configured on different entities (such as different interfaces), should not conflict—while both minimizing implementation complexity and maximizing performance of such configuration changes. Certain previous approaches to merging configurations do not satisfy such goals (among others). In particular, merging configurations based on the underlying (i.e., binary) data structures used to store the configuration of such features in memory is both complex and relatively non-performant, while using a standard text based merge may not adequately address or resolve conflicts that occur (i.e., may not minimize conflicts).

(19) To address those desires, among others, embodiments may perform a merge between an ancestor configuration, a session configuration and a current configuration of a network device based on save blocks. These save blocks may be a structured configuration model representing features to be configured. In this manner, by utilizing these save blocks as the basis for the

configurations of features and for the merging of such configurations, embodiments may generally allow such merges to be performed substantially on a per feature basis and may further allow the merging to be performed in a manner that minimizes conflicts and implementation complexity while substantially improving the performance of those configuration merges.

(20) Specifically, according to embodiments, a feature may be represented by one or more save blocks where a save block may include commands for configuring that feature. Accordingly, when a configuration session for a network device is initiated, an ancestor configuration may be obtained. In other words, the configuration of the network device at the time the configuration session is initiated may be stored. During the configuration session a user may modify the configuration (e.g., perform one or more commands against this ancestor configuration) to generate a session configuration. At some point during the configuration session, such as when a user indicates that the session configuration is to be committed, a current configuration of the network device may be obtained.

(21) Each of the ancestor configuration, session configuration and current configuration may thus be associated with save blocks (e.g., data structures defining the configuration of a feature), where some of these save blocks may include commands associated with a feature that is being configured by the user during the configuration session. The set (i.e., one or more) of save blocks for each of the ancestor configuration, session configuration and current configuration can be merged to generate a merged configuration, also comprising a set of save blocks. The network device can then be configured according to the generated merged configuration. If conflicts (e.g., between the commands or configuration associated with the feature in the ancestor configuration, session configuration or current configuration) are identified during the generation of the merged configuration these conflicts may be presented to a user (e.g., for resolution by the user). For example, the user may be prevented from committing the session configuration until the identified conflicts are resolved.

(22) In certain embodiments, the merge of these configurations may be accomplished by a three way merge algorithm applied to the save blocks of each of the ancestor configuration, session configuration and current configuration. More specifically, according to particular embodiments the save blocks of each of these configurations may be structured as a tree, such that each node of that tree is a corresponding save block. A recursive three way merge algorithm may be applied to start at the root node (save block) of each of the (tree for the) ancestor configuration, (tree for the) session configuration and (tree for the) current configuration. Accordingly, the merge may be performed between the trees on a node by node basis. In some embodiments, these save blocks may be of different types (e.g., there may be different types of save block data structures), thus the merge of a particular node of the trees of the ancestor configuration, session configuration and current configuration may be performed based on the type of save block corresponding to that node.

(23) By performing merging of the ancestor, session and current configurations, embodiments may facilitate the simple and logical configuration of network devices on a feature by feature basis while preventing the overwriting of changes to the configuration of the device and allowing the identification and resolution of conflicting configurations when they occur.

(24) Looking now at FIG. 1A, a computing system including one embodiment of a network device adapted for merging configurations is depicted. The system includes a network device **100** connected to one or more external entities (e.g., **102A**, **102N**). The network device **100** and the external entities **102** may be directly or indirectly connected to each other using any combination of wired or wireless connections and may communicate using one or more wireless or wired protocols. In embodiments there may be other network devices (or systems) (not shown) that facilitate communication between the devices.

(25) Network device **100** may include persistent storage, memory (e.g., Random Access Memory), one or more processors (including a switch chip), and network interfaces (ports). Each port may or

may not be connected to another device on a network (e.g., a server, a switch, a router, etc.). The switch chip includes hardware for determining how to route packets at the network device **100** (e.g., to which egress port a packet should be routed). The network device **100** may thus be adapted to receive packets via its ports and process the packet to determine whether to drop or route the packet, and to which port the packet should be routed.

(26) The operation of network device **100** depends on the manner in which network device **100** is configured. For example, how the network device **100** makes the determination of whether to drop the packet, or send the packet to another device on the network may depend, in part, on whether the network element is a layer-2 (L2) switch or a layer-3 (L3) switch (also referred to as a multi-layer switch), which may perform at least some functions of a router. If the network device **100** is operating as an L2 switch, the network element may use a destination Media Access Control (MAC) address along with a forwarding table to determine out of which port to send the packet. If the network device **100** is operating as a L3 switch, the network device **100** may use the destination Internet Protocol (IP) address of a packet along with a routing table to determine out of which port to send the packet, and includes the ability to write the MAC address of the next device to receive the packet in place of its own MAC address (which the last device to send the packet wrote) in the L2 information encapsulating the packet. If the network device **100** is a multi-layer switch, the multi-layer switch includes functionality to process packets using both MAC addresses and IP addresses.

(27) External entity **102** may be, or may execute on, any type of computing system that is configured to interact with the network device **100**. For example, the external entity **102** may be a desktop computer or terminal operated by a network administrator or other type of user (e.g., may be a configuration console or the like). Users at these external network devices **102** may desire to access configuration data on features configured at the network device or otherwise manage or configure (collectively configure) network device **100** (e.g., such that network device **100** performs in a desired fashion).

(28) Embodiments of the network device **100** offer the ability for a user at an external device **102** (or multiple users at multiple external entities **102**) to perform configuration changes in, or to otherwise conduct, a configuration session. When the configuration session is committed (or at some time prior) the changes in the configuration session may be merged with the currently running configuration of the network device **100** at that time (e.g., when the configuration session is committed) and to determine when such configuration changes may conflict to provide a user (e.g., conducting the configuration session) with such information.

(29) Moving now to FIG. **1B**, an embodiment of a network device adapted for merging configurations is depicted. The network device **100** includes a data plane **112** and a control plane **114**. The data plane **112** is adapted to receive packets (ingress packets) via ports (not shown) of the network device **100**, process the packets and, as appropriate, route packets (egress packets) via the ports. The data plane **112** is also adapted to gather data plane information and to provide this data plane information to the control plane **114**. Data plane information includes, for example, network usage data, flow information based on the sampling of packets received by the data plane, information related to queue depths on ingress or egress buffers (not shown) in the data plane, other data traffic statistics or other types of data plane information.

(30) Control plane **114** is adapted to manage the operation of the network device **100** (e.g., at least partially based on the data plane information received from data plane **112**). More specifically, the control plane **114** is adapted to manage the operation of the data plane **112**. The control plane **114** includes functionality to conduct configuration sessions based on received configuration data and to provide configuration data in accordance with such configuration sessions. According to embodiments, therefore, control plane **114** may conduct configuration sessions by storing an ancestor configuration (e.g., a running configuration of the network device **100** at, or close to, the time the configuration session was initiated), and maintain a session configuration for the

configuration session by applying configuration commands received during the session to (e.g., a copy of) the ancestor configuration. The control plane **114** is thus adapted to perform a merge between the ancestor configuration and the session configuration of the configuration session and the current configuration of a network device **100** at the time the configuration session is committed. This merged configuration may be used to replace the current configuration on the network device **100** or a user may be notified that certain configuration changes may conflict such that a user may be made aware of such conflicts.

(31) Additional details about such a control plane are described with respect to the embodiment of a control plane of a network device depicted in FIG. **1C**. Here, control plane **114** may include a configuration interface (or agent) **122** and one or more system state databases **126**. The control plane **114** may also include other functionality to implement protocols, processes, or services of the network device, such as a routing information base agent, a forwarding information base agent, a simple network management protocol (SNMP) agent, etc.

(32) System state databases **126** include any single or combination of shared or non-shared data stores (e.g., memories) at one or more locations in network device **100** that store the current (running) configuration **128** of the network device (e.g., values in data structures used, accessed, stored by, etc. any of the agents or functionality of the control plane **114**) such that the running configuration **128** in system state databases **126** includes the values of the feature (e.g., variables or parameters) that are currently specified or configured for the network device (e.g., such as user names, passwords, interface definitions, addresses data, routing or forwarding table entries, or the like). The system state databases **126** store the aforementioned information in, for example, records or another data structure within the system state databases **126** (e.g. binary data structures) where these data structures may conform to a data model used by network device **100**.

(33) A user can thus initiate a configuration session **180** through configuration interface **122**. When configuration session **180** is initiated, the running configuration **128** of the network device **100** at that time is stored as an ancestor configuration **182** in a memory space of storage **108** associated with that configuration session **180** (and may be assigned a unique name that a user can subsequently refer to when interacting with the configuration session **180** through configuration interface **122**). A copy of that ancestor configuration **182** can then be used as the initial session configuration **184** for the configuration session **180**.

(34) Using the configuration interface **122** then, a user can issue configuration commands associated with the configuration session **180**. These configuration commands for the configuration session can then be applied to session configuration **184** of the configuration session. The configuration session **180** thus allows configuration changes to be made to this session configuration **184** without altering the actual running configuration **128** of the network device **100** (which, as discussed, may be altered during the time configuration session **180** is ongoing). Accordingly, configuration session **180** can be entered, modified, or exited at any time without impacting currently running system configuration **128**.

(35) At some point then, a user may commit the configuration session **180**. By committing the configuration session **180** the user is indicating that they wish to replace the running configuration **128** on the network device **100** with the session configuration **184** that includes the ancestor configuration **182** (the running configuration **128** at the time the session was initiated) plus any changes to the configuration made during the configuration session **180** (e.g., the changes in configuration caused by configuration commands issued through configuration interface **122** in association with configuration session **180**).

(36) As running configuration **128** at the time the user commits the configuration session **180** may be different from the running configuration **128** at the time the configuration session **180** was initiated (e.g., the running configuration **128** used as the ancestor configuration **182**), when configuration session **180** is committed any changes made to the running configuration **128** on the device **100** between the time the configuration session **180** was initiated and the time the

configuration session was committed may be overwritten (and thus lost).

(37) According to embodiments, therefore, configuration interface **122** may be adapted to merge configuration changes from a configuration session **180** into a currently running configuration **128** and to determine when such configuration changes may conflict to provide a user with such information. In particular, embodiments may perform a three way merge between the ancestor configuration **182** (e.g., the running configuration **128** that existed at the time configuration session **180** was initiated), the session configuration **184** of the configuration session **180** including changes to that ancestor configuration **182** during the configuration session **180**, and the current running configuration **128** existing at the time the configuration session **180** is committed. By performing this merging, embodiments may facilitate the simple and logical configuration of the network device **100** while preventing the overwriting of changes to the running configuration **128** of the device **100** and allowing the identification and resolution of conflicting configurations when they occur.

(38) FIG. 2 is a block diagram depicting a general architecture of a network device for performing merging in accordance with certain embodiments. Network device **200** may be a router, switch, server, or any other computing device that may be configured to control or process network traffic. The network device **200** may receive data, including packets from hosts (not shown), via an input/output (I/O) path **202**. I/O path **202** may provide packet data to control circuitry **204**, which includes processing circuitry **206** and storage (i.e., memory) **208**. Control circuitry **204** may send and receive commands, requests, and other suitable data using I/O path **202**. I/O path **202** may connect control circuitry **204** (and specifically processing circuitry **206**) to one or more network interfaces **212** to which other devices of a network (e.g., hosts) can be connected. These network interfaces **212** may be any type of network interface, such as an RJ45 ethernet port, a coaxial port, or a serial port such as RS232, etc.

(39) Control circuitry **204** includes processing circuitry **206** and storage **208**. As referred to herein, processing circuitry should be understood to mean circuitry based on one or more microprocessors, microcontrollers, digital signal processors, programmable logic devices, field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), etc., and may include a multi-core processor (e.g., dual-core, quad-core, hexa-core, octa-core, or any suitable number of cores). In some embodiments, processing circuitry **206** is distributed across multiple separate processors or processing units, for example, multiple of the same type of processing units (e.g., two INTEL CORE i7 processors) or multiple different processors (e.g., an INTEL CORE i5 processor and an INTEL CORE i7 processor). The circuitry described herein may execute instructions included in software running on one or more general purpose or specialized processors.

(40) Storage **208** may be an electronic storage device that includes volatile random-access memory (RAM) **230**, which does not retain its contents when power is turned off, and non-volatile RAM **232**, which does retain its contents when power is turned off. As referred to herein, the phrase “memory”, “electronic storage device,” or “storage device” should be understood to mean any device for storing electronic data, computer software, instructions, or firmware, such as RAM, content-addressable memory (CAM) (including a TCAM), hard drives, optical drives, solid state devices, quantum storage devices, or any other suitable fixed or removable storage devices, or any combination of the same. The running configuration of the device **200** may be stored in storage **208**. This running configuration may include values specified or otherwise configured for one or more features of network device **200**. For example, storage **208** may include system state databases that include data structures (e.g. binary data structures) comprising the values for these features.

(41) Control circuitry **204** executes instructions for conducting configuration sessions for configuring the network device **200** and stores these configuration sessions, including an associated ancestor configuration and session configuration in storage **208**. As the configuration session is conducted, the control circuitry **204** (e.g., instructions executing thereon) can execute received configuration commands for the configuration session against the session configuration to update

the session configuration. The configuration commands can, for example, be adapted to configure values for certain features of the network device **200**.

(42) When the configuration session is committed, the control circuitry **204** is adapted to perform a merge between an ancestor configuration, a session configuration, and a current configuration of a network device. In one embodiment, such a merge is performed based on a data structure associated with a configuration model (referred to as a save block) where each of the save blocks represents (e.g., defines) the configuration of one or more features of the network device **200**. Specifically, a save block may include a list of commands that define a configuration for the associated one or more features. A set of save blocks can thus be generated for the ancestor configuration, the session configuration, and the current running configuration of the network device **200**.

(43) A merged configuration can then be generated for the configuration session by the control circuitry **204** by performing a three way merge between the ancestor configuration, the session configuration, and the current running configuration using the generated save blocks for each of the ancestor configuration, the session configuration, and the current running configuration. The network device **200** can then be configured according to the generated merged configuration. If conflicts (e.g., between the commands or configuration associated with the feature in the ancestor configuration, session configuration or current configuration) are identified by control circuitry **204** during the generation of the merged configuration these conflicts may be presented to a user (e.g., for resolution by the user). For example, the user may be prevented from committing the session configuration until the identified conflicts are resolved.

(44) FIG. 3 is a more detailed depiction of an embodiment of a network system including a network device **300** and an external device such as a configuration console **360** coupled to the network device **300** over a communication network **390**, where device **300** merges configurations when committing a configuration session. Specifically, configuration interface **358** (executing on processing circuitry **306** included in control circuitry **304**) may be adapted to conduct a configuration session. Thus, a user at configuration console **360** thus initiates a configuration session **380** through configuration interface **358**. When configuration session **380** is initiated, the running configuration **376** of the network device **300** at that time (t1) is stored as an ancestor configuration **374** in a memory space of storage **308** associated with that configuration session **380** (and may be assigned a unique name that a user can subsequently refer to when interacting with the configuration session **380** through configuration interface **358**). A copy of that ancestor configuration **374** (e.g., running configuration from time t1) can then be used as the initial session configuration **372** for the configuration session **380**.

(45) Using the configuration interface **358** then, a user can issue configuration commands (e.g., configuration data) associated with the configuration session **380**. These configuration commands for the configuration session can then be applied to session configuration **380** of the configuration session by the configuration interface **358**. The configuration session **380** thus allows configuration changes to be made to this session configuration **372** without altering the actual running configuration **376** of the network device **300** (which may be altered during the time configuration session **380** is ongoing). Accordingly, configuration session **380** can be entered, modified, or exited by the user at the configuration console **360** at any time without impacting currently running system configuration **376**.

(46) At some point then (e.g., time t2), a user at the configuration console **360** may commit the configuration session **380**. By committing the configuration session **380** the user is indicating that they wish to replace the running configuration **376** on the network device **300** with the session configuration **372** that includes the ancestor configuration **374** (the running configuration **376** at the time (t1) the session **380** was initiated) plus any changes to this configuration made during the configuration session **380** by a user issuing configuration commands at configuration console **360**.

(47) As running configuration **376** at the time (t2) the user commits the configuration session **380**

may be different from the running configuration **376** at the time (t1) the configuration session **380** was initiated, configuration interface **358** may be adapted to merge configuration changes from a configuration session **380** into the currently running configuration **376** and to determine when such configuration changes may conflict to provide a user with such information. According to an embodiment, configuration interface **358** may therefore include configuration merger **354**.

(48) Configuration merger **354** is adapted to merge configurations associated with a configuration session **380** by performing a three way merge between the ancestor configuration **374** for the configuration session **380** (e.g., the running configuration **376** that existed at the time (t1) the configuration session **380** was initiated), the session configuration **372** of the configuration session **380** including changes to that ancestor configuration **374** during the configuration session **380**, and the current running configuration **376** existing at the time (t2) the configuration session **380** is committed.

(49) In particular, according to one embodiment, when the user commits the configuration session **380** (e.g., at time (t2)), the configuration merger **354** may obtain the session configuration **372** and the ancestor configuration **374** associated with the committed configuration session **380**. Additionally, the configuration merger **354** can obtain the current running configuration **376** at that time (e.g., time t2 when the session is committed).

(50) The configuration merger **354** can then generate save blocks for each of the ancestor configuration **374** (from time t1), the session configuration **372** and the currently running configuration **376** (from time t2). These save blocks may include a list of commands or other definitions that define or otherwise represent a configuration for one or more features of the network device **300**. To generate the save blocks for a configuration the configuration merger **354** may, for example, evaluate the (e.g., binary) data structures including the configurations for the features as included in that configuration and generate the corresponding save blocks representing those configurations for those features. These save blocks can, for example, be formed according to a structured configuration model and include commands for those features adapted to configure (or represent the configuration of) those features as defined in that configuration. Accordingly, a corresponding set of save blocks can be generated for each of the ancestor configuration **374**, the session configuration **372** and the currently running configuration **376**.

(51) Once configuration merger **354** generates the corresponding set of save blocks for each of the ancestor configuration **374**, the session configuration **372** and the currently running configuration **376**, the configuration merger **354** can merge the generated set of save blocks for each of the session configuration, the current configuration, and ancestor configuration to generate merged configuration **378**. This merge can be accomplished using a merging algorithm such as a three way merge or the like. The resulting merged configuration **378** may therefore include a set of save blocks determined based on the merging of each of the set of save blocks for the session configuration **372**, the current running configuration **376**, and the ancestor configuration **374**, where the save blocks of merged configuration **378** may include commands for those features adapted to configure (or represent the configuration of) those features resulting from such a merge. In some embodiments, the merge of the session configuration **372**, current running configuration **376**, and ancestor configuration **374** may be accomplished by applying the merge algorithm on a save block by save block basis, such that corresponding save blocks from each of the set of save blocks for the session configuration **372**, current running configuration **376**, and ancestor configuration **374** are merged to generate a corresponding save block of the set of save blocks for the merged configuration **378**.

(52) Once the merged configuration **378** is generated it can then be used to configure the network device (e.g., by replacing the running configuration **376** with the merged configuration **378** at time t3). In some embodiments, during the merge of the session configuration **372**, current running configuration **376**, and the ancestor configuration **374** the configuration merger **354** may determine any conflicts that may exist between each of these configurations. These conflicts may be identified

to a user (e.g., through configuration console **360**) such that a user may resolve such conflicts or take other action. For example, the merging of the configuration may be paused or the user may be prevented from committing the configuration session **380**.

(53) FIGS. **4A** and **4B** are a detailed depiction of an embodiment of a method for performing such a merge for a configuration session to generate a merged configuration and the application of a merged configuration to the running configuration of a network device. As discussed, a configuration session may be based on an ancestor configuration comprising the running configuration of the network device at the time the configuration session was initiated. During the configuration session a session configuration that includes the ancestor configuration modified according to any configuration changes made during the configuration session is generated. Accordingly, when a configuration session is committed (or at some other point where it is desired to merge configurations), this session configuration may be merged with the ancestor configuration and a current running configuration of the network device.

(54) In one embodiment, then, the ancestor configuration **474** and session configuration **472** of the configuration session may be obtained, along with a current running configuration **476** of the network device, and a corresponding set of save blocks **450** for each of those configurations generated (STEP **402**). Each of the save blocks **450** can thus include a list of commands or other definitions that define or otherwise represent a configuration for one or more features of the network device. To generate the save blocks for a configuration, a (e.g., binary) data structure including the configuration for a feature can be evaluated and a corresponding save block **450** representing the configurations for that feature generated. These save blocks **450**, can, for example be formed according to a structured configuration model and include commands for those features adapted to configure (or represent the configuration of) those features as defined in that configuration. Moreover, these save blocks **450** may be orthogonal to one another (for a particular configuration) in certain embodiments, such that save blocks **450** of a set of save blocks for a particular configuration (e.g., ancestor configuration **474**, session configuration **472** and current running configuration **476**) may not overlap (e.g., may not include commands for the same feature, or define the same parameter or value for the same feature).

(55) In some embodiments, the structured configuration model for save blocks **450** may define different types of save blocks. For example, in one embodiment there may be three types of save blocks defined, a “CommandSequence” save block, a “ModeCollection” save block and a “Mode” save block. A CommandSequence save block may include a set of commands associated with the configuration of a feature. ModeCollection and Mode save blocks may be save blocks for structuring, grouping or otherwise organizing save blocks. A ModeCollection may include (or references) a set of Mode save blocks, where all of the referenced Mode save blocks may be of the same type of mode. For example, each of the referenced save blocks may be children of the node comprising a ModeCollection save block in the case where a set of save blocks are structured as a tree. A Mode save block includes (or references) other save blocks of type CommandSequence or ModeCollection.

(56) It may be useful to here describe examples of save blocks. Suppose for example, an ancestor configuration for a feature is: interface Ethernet1 description foo

(57) A session configuration for the same feature is: interface Ethernet1 description foo no switchport ip address 10.1.1.1/24

(58) And a current running configuration is: interface Ethernet1 description bar

(59) In this example, the corresponding save blocks for the ancestor configuration would be: GlobalConfigMode Mode ‘interface Ethernet1’ CommandSequence ‘Arnet.intf’->[‘description foo’]

(60) The corresponding save blocks for the session configuration would be: GlobalConfigMode Mode ‘interface Ethernet1’ CommandSequence ‘Arnet.intf’->[‘description foo’]
CommandSequence ‘Ebra.switchport’->[‘no switchport’] CommandSequence ‘Ira.ipIntf’->[‘ip

address 10.1.1.1/24']

(61) And the corresponding save blocks for the running configuration would be:

GlobalConfigMode Mode 'interface Ethernet1' CommandSequence 'Arnet.intf'->['description bar']

(62) Accordingly, a corresponding set of save blocks **450** can be generated for each of the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476**. In one embodiment, each of the sets of save blocks **450** for each of the session configuration **472**, current configuration **476**, and ancestor configuration **474** are structured as a tree, such that each node of the tree comprises a corresponding one of the set of save blocks **450** where each interior node of the tree has one or more child nodes that are also save blocks **450**. Thus, each of the set of save blocks **450** for each of the configurations **472**, **474**, **476** may be structured as an identical, or substantially similar, tree structure with corresponding nodes (e.g., save blocks **450** that may pertain to the same feature or set of features or save blocks **450** associated with the same structuring, grouping or organization save blocks). For example, root node save block **450a1** of ancestor configuration **474** may correspond to root node save block **450b1** of current configuration **476** and root node save block **450c1** of session configuration **472**; interior node save block **450a2** of ancestor configuration **474** may correspond to interior node save block **450b2** of current configuration **476** and interior node save block **450c2** of session configuration **472**; leaf node save block **450a3** of ancestor configuration **474** may correspond to leaf node save block **450b3** of current configuration **476** and leaf node save block **450c3** of session configuration **472**, etc.

(63) Once the corresponding set of save blocks **450** are generated for each of the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476**, these generated sets of save blocks **450** for each of the configurations **472**, **474**, **476** (e.g., structured as a tree) can be merged to generate a merged configuration **478** (STEP **404**). The merge may be accomplished by generating differences between the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476** at the save block level.

(64) In one embodiment, this merge can be accomplished using a merging algorithm such as a three way merge or the like. For example, in cases where the sets of save blocks **450** for each of the configurations are structured as a tree, a recursive three way merger may be performed on the save blocks **450** of the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476** starting at the root node (e.g., save blocks **450a1**, **450b1**, **450c1**) of the each of the trees of save blocks for each of the configurations **472**, **474**, **476** and recursing through the trees of save blocks **450** for each configuration to compare the corresponding save blocks **450** for each of the configurations.

(65) Thus, the merge of the set of save blocks **450** (e.g., structured as a tree) may be performed on a node by node basis by comparing the corresponding save blocks **450** in each of the set of save blocks for each of the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476** (e.g., save blocks **450a2**, **450b2** and **450c2** may be compared, save blocks **450a3**, **450b3**, **450c3** may be compared to one another). This merged configuration **478** may thus comprise a set of save blocks **450**, each of the save blocks **450** of the merged configuration **478** corresponding to (and resulting from the merge of) corresponding save blocks **450** of the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476**.

(66) In some embodiments, it can be determined if there are any conflicts when merging save blocks **450** from the different configurations **472**, **474**, **476** (STEP **406**). If such conflicts are identified they may be returned to the user (Y branch of STEP **406** and STEP **408**), and in some embodiments the merge of the configurations **472**, **474**, **476** may be stopped, or the committing of the session configuration **472** otherwise halted. If no conflicts are identified the merged configuration **478** may be generated (N branch of STEP **406**).

(67) Such conflicts may arise in circumstances where, for example, a corresponding save block **450**

from the session configuration **472** and the currently running configuration **476** both differ from the corresponding save block **450** of the ancestor configuration **474** (e.g., a command or value for a feature differs between the session configuration **472** and the ancestor configuration **474** and the command or value for that same feature differs between the currently running configuration **476** and the ancestor configuration **474**). Other conflicts may arise if save blocks **450** or collections of save blocks **450** are present (or not present) or different between the configurations **472**, **474**, **476**. It will be noted here that these examples of conflicts are given without loss of generality and there may be additional, fewer or different conflicts that may for example, be defined with respect to the particular embodiment of merge performed, or specified or configured by a user.

(68) In certain embodiments, the comparison between corresponding save blocks **450** of the ancestor configuration **474**, the session configuration **472** and the currently running configuration **476** may be performed based on the type of the save block **450** being compared. For example, when merging save blocks **450** from the configurations **472**, **474**, **476** if the corresponding save blocks **450** from the configurations **472**, **474**, **476** being compared are of the CommandSequence type the set of commands for the feature in the save block **450** from the session configuration **472** can be compared to the set of commands for the feature of the corresponding save block **450** of the current running configuration **476**. If the set of commands for the features from these save blocks **450** are different, the set of commands for the feature of save block **450** of session configuration **472** may be compared to the corresponding save block **450** from the ancestor configuration **474**, and the set of commands for the feature of save blocks **450** of current running configuration **476** may also be compared to the corresponding save block **450** from the ancestor configuration **474**.

(69) If both the (set of commands of) save block **450** of session configuration **472** and the (set of commands of) save block **450** of current running configuration **476** are different from the (set of commands of) corresponding save block **450** of the ancestor configuration **474** this can be identified as a conflict (e.g., and identified to a user). Otherwise, the set of commands for the feature (e.g., in save block **450** of session configuration **472** or save block **450** of current running configuration **476**) that differ from the corresponding save block **450** of the ancestor configuration **474** may be obtained and used to form corresponding save block **450** of merged configuration **478**. If there are no such differences the set of commands for the feature from the save block **450** of the session configuration **472** may be used to form corresponding save block **450** of merged configuration **478**.

(70) As another example, recall that a ModeCollection type of save block may be for structuring, grouping or otherwise organizing save blocks **450** and may thus include or reference such save blocks **450**. Thus, a comparison of corresponding ModeCollection save blocks may be based on the save blocks **450** included or referenced in the corresponding ModeCollection save blocks **450** being compared. For example, if a ModeCollection save block **450** of the running configuration **476** (e.g., the save block being compared or a save block **450** referenced or included in a ModeCollection save block being compared) is not present in the save blocks **450** of the session configuration **472**, it can be determined if that save block **450** is present in the save blocks **450** of the ancestor configuration **474**. If that save block **450** is present in the ancestor configuration **474**, and the save block **450** of the running configuration **476** is not the same as that corresponding save block **450** of the ancestor configuration **474**, a conflict can be identified. In cases where that save block **450** of the running configuration **476** and the ancestor configuration **474** are the same, it may not be added to the save blocks of the merged configuration **478**.

(71) Similarly, when a ModeCollection save block **450** is present in save blocks **450** of the session configuration **472** it can be determined if that save block **450** is present in save blocks **450** of the running configuration **476**. If that save block **450** is not present in save blocks **450** of the running configuration **476** it can be determined if a corresponding save block **450** is present in the ancestor configuration **474**. If that save block **450** is present in the ancestor configuration **474**, and the save block **450** of the session configuration **472** is not the same as that corresponding save block **450**

present in the ancestor configuration **474**, a conflict can be identified. Here, in cases where save block **450** of the session configuration **472** and the ancestor configuration **474** are the same, it may be added to the save blocks **450** of the merged configuration **478**.

(72) If a ModeCollection save block **450** is present in both the save blocks **450** of the session configuration **472** and the save blocks **450** of the running configuration **476** it can be determined if that save block **450** is present in the save blocks **450** of the ancestor configuration **474**. If the corresponding ModeCollection save block **450** is present in the ancestor configuration **474**, a recursive merge can continue on each of the save blocks referenced by that ModeCollection save block **450**. If, however, a corresponding ModeCollection save block **450** is not present in the ancestor configuration **474**, this may indicate that the save block **450** is new. In this case, if the corresponding ModeCollection save block **450** is different between the session configuration **472** and the running configuration **476** a conflict can be identified. Otherwise, that ModeCollection save block **450** of the session configuration **472** may be included in the save blocks **450** of the merged configuration **478**.

(73) In some embodiments, corresponding Mode save blocks **450** in different configurations **472**, **474**, **476** may be considered the same if one or more of the same commands or comment, or a particular type of command or comment of the corresponding Mode save blocks **450** is the same. For example, two Mode save blocks **450** may be considered the same if they have the same commands (e.g., an “enter” command) associated with a same ModeCollection save block. In some embodiments, a Mode save block **450** may include a comment or description, and reference a set of save blocks **450** (e.g., a ModeCollection save block). Thus, in one embodiment, to merge a corresponding Mode save block **450** of session configuration **472**, running configuration **476** and ancestor configuration **474** to determine a corresponding save block **450** of the merged configuration **478**, the comment portions of the corresponding Mode save blocks **450** can be merged to determine a comment portion of the resulting save block to be included in the merged configuration **478**. Additionally, the reference portion (e.g., the set of referenced save blocks) of the corresponding Mode save blocks **450** can also be merged to determine the set of save blocks **450** for the referenced save blocks of the resulting Mode save block **450** to be included in the merged configuration **478**. The merge of the corresponding Mode blocks **450** referenced save blocks **450** may be accomplished substantially as described above with respect to the merging of save blocks in a ModeCollection save block.

(74) A merge for a comment portion of a Mode save block **450** may be determined in a similar manner to that described with respect to a CommandSequence save block **450**. Specifically, if the comment portion of the Mode save block **450** from the session configuration **472** is different than the comment portion of the Mode save block **450** of the running configuration **476**, the comment portion of the corresponding save block **450** of the ancestor configuration **474** may be utilized to determine which one of the save block **450** from the session configuration **472** or the save block **450** from the running configuration **476** includes the different comment portion (or if both of those save blocks **450** include a changed comment portion). If the comment portion of both the save blocks **450** from the session configuration **472** and the running configuration **476** changed relative to the ancestor configuration **474** then a conflict may be identified. Otherwise, the comment portion of the save block **450** of the running configuration **476** or the session configuration **472** that changed relative to the ancestor configuration **474** may be used to form the corresponding save block **450** of the merged configuration. If the comment portions of all the corresponding save blocks **450** from the session configuration **472**, ancestor configuration **474** and running configuration **476** are all the same then the comment portion of the save block **450** of the session configuration **472** may be utilized to form the corresponding save block of the merged configuration **478**.

(75) To illustrate a merge with respect to the above example, the difference (e.g., any additions or modifications) between the save block for the ancestor configuration and the session configuration

may be: GlobalConfigMode Mode 'interface Ethernet1' CommandSequence 'Ebra.switchport'->['no switchport'] CommandSequence 'Ira.ipIntf'->['ip address 10.1.1.1/24']

(76) These determined differences can be applied to the running configuration by merging these determined differences with the (e.g., corresponding save block of) the running configuration. In this case then, the application (merging) of the determined differences (between the save block for the ancestor configuration and the session configuration) and the (save block of the) running configuration yields the following, which may be considered a merged save block:

GlobalConfigMode Mode 'interface Ethernet1' CommandSequence 'Arnet.intf'->['description bar'] CommandSequence 'Ebra.switchport'->['no switchport'] CommandSequence 'Ira.ipIntf'->['ip address 10.1.1.1/24']

(77) Thus, once the sets of save blocks **450** for each of the configurations session session configuration **472**, ancestor configuration **474** and current running configuration **476** are merged to generate a merged configuration **478**, the session configuration **472a** may be rolled back to generate a rolled back session configuration **472b** (STEP **410**). Such a rolled back session configuration **472b** may be substantially similar to ancestor configuration **474**. In one embodiment, only the save blocks **450** (e.g., nodes or roots of the tree structure of session configuration **472a**) that have been modified during the configuration session may be returned to the state at the beginning of the configuration session (e.g., to correspond with ancestor configuration **474**).

(78) The determined merged configuration **478** can then be applied to the rolled back session configuration **472b** (STEP **412**) by merging the save blocks **450** of the merged configuration **478** with the corresponding save blocks **450** of the rolled back session configuration **472b**. The resulting session configuration **472c** (e.g., which comprises the merged configuration **478**) can then be used to configure the network device by replacing the (current) running configuration of the network device with that session configuration **472c** (STEP **414**).

(79) It will be understood that while specific embodiments have been presented herein, these embodiments are merely illustrative, and not restrictive. Rather, the description is intended to describe illustrative embodiments, features and functions in order to provide an understanding of the embodiments without limiting the disclosure to any particularly described embodiment, feature, or function, including any such embodiment, feature, or function described. While specific embodiments of, and examples for, the embodiments are described herein for illustrative purposes only, various equivalent modifications are possible within the spirit and scope of the invention, as those skilled in the relevant art will recognize and appreciate.

(80) As indicated, these modifications may be made in light of the foregoing description of illustrated embodiments and are to be included within the spirit and scope of the disclosure. Thus, while particular embodiments are described, a latitude of modification, various changes and substitutions are intended in the foregoing disclosures, and it will be appreciated that in some instances some features of embodiments of the invention will be employed without a corresponding use of other features, and features described with respect to one embodiment may be combined with features of other embodiments without departing from the scope and spirit of the disclosure as set forth.

Claims

1. A method for configuring a network device, comprising: obtaining an ancestor configuration of a network device, wherein a configuration session was initiated using the ancestor configuration and the ancestor configuration is associated with a first time; obtaining a session configuration of the network device, wherein the session configuration is based on the ancestor configuration and is configured during the configuration session; obtaining a current configuration of the network device associated with a second time; generating a set of save blocks for each one of the session configuration, the current configuration, and the ancestor configuration, wherein each of the

session configuration, current configuration and ancestor configuration define structured configuration models associated with features of the network device and generating the set of save blocks for each configuration comprises evaluating the structured configuration model of that configuration to form the set of save blocks representing configurations for the features defined by that configuration such that at least one of the set of save blocks generated for the session configuration comprises a first command for configuring a feature on the network device, at least one of the set of save blocks generated for the current configuration comprises a second command for configuring the feature on the network device, and at least one of the set of save blocks generated for the ancestor configuration comprises a third command for configuring the feature on the network device; merging the set of save blocks for the session configuration, the set of save blocks for the current configuration, and the set of save blocks for the ancestor configuration based on a comparison of the set of save blocks for the session configuration, the set of save blocks for the current configuration, and the set of save blocks for the ancestor configuration to generate a merged configuration comprising a set of save blocks, wherein the merging is based on the first command, second command, and third command for configuring the feature, and the set of save blocks of the merged configuration comprises at least one save block comprising a command for configuring the feature on the network device; and configuring the network device according to the merged configuration by altering the current configuration of the network device based on the command of the at least one save block of the merged configuration.

2. The method of claim 1, further comprising: identifying a conflict between the command associated with the feature in the ancestor configuration, the session configuration, or the current configuration based on the generation of the merged configuration; and presenting the conflict to a user.

3. The method of claim 1, wherein the set of save blocks for each of the session configuration, the current configuration, and the ancestor configuration are structured as a tree such that each node of the tree comprises a corresponding one of the set of save blocks, and the merge is a recursive merge starting with a root node of each tree.

4. The method of claim 3, wherein the merge is a three way merge.

5. The method of claim 3, wherein the merge of each of the set of save blocks for the session configuration, the current configuration, and the ancestor configuration is performed on a node by node basis.

6. The method of claim 5, wherein the merge of each node of the tree is based on a type of the save block corresponding to that node.

7. The method of claim 1, wherein the second time corresponds to a commit of the configuration session.

8. The method of claim 1, wherein configuring the network device according to the merged configuration comprises rolling back the session configuration.

9. The method of claim 8, wherein configuring the network device according to the merged configuration further comprises: merging the merged configuration with the rolled back session configuration; and overwriting the current configuration at the network device with the session configuration.

10. A configuration system, comprising: a processor; a data store storing an ancestor configuration of a network device from a first time associated with an initiation of a configuration session, a current configuration from a second time associated with a commit of the configuration session, and a session configuration defined during the configuration session, wherein each of the ancestor configuration, current configuration and session configuration define structured configuration models associated with features of the network device, and wherein the ancestor configuration comprises a first tree, each node of the first tree comprising a save block defined based the structured configuration model associated with the ancestor configuration and at least one save block of the first tree comprises a first command associated with a feature, the current

configuration comprises a second tree defined based on the structured configuration model associated with current configuration, each node of the second tree comprising a save block and at least one save block of the second tree comprises a second command associated with the feature, and the session configuration comprises a third tree defined based on the structured configuration model associated with the session configuration, each node of the third tree comprising a save block and at least one save block of the third tree comprises a third command associated with the feature; and a non-transitory computer readable medium, comprising instructions for: generating a merged configuration comprising a fourth tree, each node of the fourth tree comprising a save block determined by merging the ancestor configuration, the current configuration and the session configuration, and at least one save block of the fourth tree comprises a command associated with the feature wherein the merging comprises merging the first command, second command, and third command associated with the feature, and wherein merging the ancestor configuration, the current configuration and the session configuration comprises comparing save blocks of the ancestor configuration, current configuration and session configuration by applying a recursive three way merge to merge save blocks of the first tree of the ancestor configuration, save blocks of the second tree of the current configuration and save blocks of the third tree of the session configuration on a node by node basis; and configuring the network device according to the merged configuration when no conflict is identified while merging the ancestor configuration, the current configuration, and the session configuration, wherein configuring the network device comprises altering the current configuration of the network device based on at least one save block of the merged configuration.

11. The configuration system of claim 10, wherein the instructions are further for presenting a conflict to a user when the conflict is identified while merging the ancestor configuration, the current configuration, and the session configuration.

12. The configuration system of claim 10, wherein the instructions are further for: rolling back the session configuration; merging the merged configuration with the session configuration; and configuring the network device according to the rolled back session configuration to configure the network device according to the merged configuration.

13. The configuration system of claim 10, wherein the merge of each node of the first tree, second tree and third tree is based on a type of save block corresponding to that node.

14. The configuration system of claim 10, wherein the merged configuration is generated in response to a commit of the configuration session.

15. A non-transitory computer readable medium, comprising instructions for: obtaining an ancestor configuration of a network device associated with a configuration session initiated at a first time; obtaining a session configuration associated with the configuration session, wherein the session configuration is based on the ancestor configuration and is configured during the configuration session; obtaining a current configuration of the network device associated with a second time; generating a set of save blocks for each one of the session configuration, the current configuration, and the ancestor configuration, wherein each of the session configuration, current configuration and ancestor configuration define structured configuration models associated with features of the network device and generating the set of save blocks for each configuration comprises evaluating the structured configuration model of that configuration to form the set of save blocks representing configurations for the features defined by that configuration such that at least one of the set of save blocks generated for the session configuration comprises a first command for a first configuration or a second configuration of a feature of the network device, at least one of the set of save blocks generated for the current configuration comprises a second command for the first configuration or the second configuration of the feature of the network device, and at least one of the set of save blocks generated for the ancestor configuration comprises a third command for the first configuration or the second configuration of the feature of the network device; merging the set of save blocks for the session configuration, the set of save blocks for the current configuration, and

the set of save blocks for the ancestor configuration based on a comparison of the set of save blocks for the session configuration, the set of save blocks for the current configuration, and the set of save blocks for the ancestor configuration to generate a merged configuration comprising a set of save blocks, wherein the merging is based on the first command, second command, and third command and the set of save blocks of the merged configuration comprises either the first configuration or the second configuration of the feature on the network device; and configuring the network device according to the merged configuration when no conflict is identified while merging the set of save blocks for each of the session configuration, the current configuration, and the ancestor configuration by altering the current configuration of the network device based on at least one save block of the merged configuration.

16. The non-transitory computer readable medium of claim 15, wherein the second configuration is a different configuration for the feature than the first configuration.

17. The non-transitory computer readable medium of claim 15, where the merging of the set of save blocks is a recursive three way merge.

18. The non-transitory computer readable medium of claim 15, wherein merging the set of save blocks for each of the session configuration, the current configuration, and the ancestor configuration comprises merging corresponding save blocks from each of the set of save blocks for the session configuration, the current configuration, and the ancestor configuration.

19. The non-transitory computer readable medium of claim 18, wherein a corresponding save block from the session configuration, the current configuration, and the ancestor configuration comprises a command associated with the first configuration or the second configuration for the feature.

20. The non-transitory computer readable medium of claim 18, further comprising instructions for identifying a conflict based on the first configuration and the second configuration in the ancestor configuration, the session configuration, or the current configuration when merging the sets of save blocks for each of the session configuration, the current configuration, and the ancestor configuration.
